Bayesian Black-Box Optimisation

Project FAQ

Author: Erkan Keskin - Neuroxa-Labs

https://github.com/Neuroxa-Labs/Bayesian_Optimisation

Companion to docs/final_report.pdf

Version: v2 (post-Week-13 portal results)

Imperial College London - PCMLAI Stage 2 - Neuroxa-Labs

Imperial College London - PCMLAI Stage 2 capstone Author: Erkan Keskin - Organisation: Neuroxa-Labs Repository: https://github.com/Neuroxa-Labs/Bayesian_Optimisation

This FAQ answers common questions about this repository and how the weekly black-box challenge was run. It is a portfolio companion to the final research report, not a substitute for the official course FAQ / brief.

A. Challenge basics

A1. What is the task?

Maximise eight unknown continuous black-box functions f_i : [0,1]^{d_i} -> R (dimensions 2 through 8). Each week you may submit one query per function. The portal returns a scalar y. Higher y is always better. The true formulas are withheld.

A2. What counts as a "query"?

A six-decimal portal string such as 0.440000-0.980000-0.980000-0.980000. Coordinates must lie in [0,1]. After the portal returns y, that pair is appended to the function's history and used for the next decision.

A3. Do we need to recover the formula?

No. The goal is sample-efficient sequential decisions that find high-performing x values under a tiny weekly budget - not symbolic identification.

A4. How many observations do we have?

Each function starts with an initial design (seed points in data/function_/). Weekly portal points are appended. After Week 13, typical totals are roughly 23-53 points depending on dimension and seed size (see the final report table).

B. Method

B1. What is the primary optimiser?

A Gaussian Process with a Matérn kernel and ARD length scales, scored with Expected Improvement (EI) or Upper Confidence Bound (UCB), plus a trust region around the incumbent for late exploit.

B2. Why not a neural net as the weekly decider?

With n ~ 20-50 points, calibrated uncertainty matters more than flexible curve-fitting. Neural nets / large ensembles were used by some peers as diagnostics; they were not our primary portal decider.

B3. What is the F1 "trust gate"?

Most of the F1 map reads ~0 (near-null labels). Exploiting a GP on a null map wastes budget on boundary artefacts. The trust gate refuses aggressive exploit until a measurable signal cluster appears; Weeks 10-12 opened a lobe near (0.64, 0.68).

B4. Why hard-return on F2 and F6?

Both have sharp basins. A "nearby" neighbour step can collapse y (F2 ~0.54 vs peak ~0.78; F6 Week-11 collapse after leaving the Week-10 centroid). Late policy returns toward the historical best after a failed neighbour.

B5. What changed in Weeks 10-13?

Near-pure exploitation of proven regions: shrink trust regions, move only ARD-sensitive axes, hard-return after misses, and (F5) continue the ridge climb on x_1.

C. Results (after Week 13)

C1. What are the best-so-far values?

- Fn	Best y	Status note
- F1	7.711x10^-16	Unresolved; late lobe to -0.00457
- F2	0.776645	Strong sharp ridge (W13 miss)
- F3	-0.011366	Safe local band held
- F4	0.679389	Strong late climb (W13)
- F5	3812.75	Clearest ridge success
- F6	-0.136	W10 basin; fragile
- F7	1.877841	Strong late climb (W13)
- F8	9.873669	Strong late climb (W13)

C2. Did late weeks still improve?

Yes. Improve counts: W8 3/8 - W9 4/8 - W10 5/8 - W11 4/8 - W12 4/8 - W13 4/8 (F4, F5, F7, F8).

C3. Is F1 "solved"?

Not in this repository. Seed max remains the official best. Peers who found a large positive peak (e.g. ~0.5-2) demonstrate that a true spike exists; our Weeks 10-12 lobe is the first usable local basin we observed.

C4. What happened in Week 13?

4/8 improved (F4, F5, F7, F8). F5 reached 3812.75. F2/F6 missed historical basins; F1 lobe continued; F3 held. See weeks/WEEK13_REFLECTION.md (../weeks/WEEK13_REFLECTION.md).

D. Repository map

D0. Where is the strategy hub?

Root STRATEGY.md (../STRATEGY.md) - per-function GP/AF parameters, locks, late policy, and links into weekly notes.

D1. Where is the executable code?

notebooks/BBO_Capstone_Optimized.ipynb (../notebooks/BBO_Capstone_Optimized.ipynb)

D2. Where is the written evidence pack?

- Markdown: docs/final_report.md (final_report.md)
- PDF: docs/final_report.pdf (final_report.pdf)
- This FAQ (PDF): docs/project_faq.pdf (project_faq.pdf)

D3. Where are figures and progress dashboards?

- Gallery: reports/analysis/README.md (../reports/analysis/README.md)
- Progress: reports/progress/README.md (../reports/progress/README.md)
- Per-function plots: data/function_/analysis_F.png

D4. Where are transparency artefacts?

DATASHEET.md (../DATASHEET.md) - MODEL_CARD.md (../MODEL_CARD.md) - docs/TECHNICAL_JUSTIFICATION.md (TECHNICAL_JUSTIFICATION.md)

D5. How do I navigate course-style deliverables?

docs/COURSE_INDEX.md (COURSE_INDEX.md)

E. Practical / reproducibility

E1. Can I reproduce a weekly decision?

Yes in spirit: reload that week's history from data/function_/, fit the GP with the notebook settings, and re-score candidates. Exact floating-point portal strings are logged in weekly strategy notes under weeks/.

E2. Are random seeds fixed?

GP multi-restart fits and candidate searches can retain stochasticity. Treat weekly notes + locked portal strings as the audit trail for what was actually submitted.

E3. What should I read first as a new visitor?

- Root README.md (../README.md)
- This FAQ or final_report.pdf (final_report.pdf)
- reports/analysis/README.md (../reports/analysis/README.md) for visuals

F. Honest limitations

- One query per function per week - no inner real-function line search.
- GPs can be confidently wrong in empty regions (inflated ? -> boundary chase).
- Clustered sampling can miss a distant second mode (especially F1).
- F2's observed peak may be an optimistic draw under noise.
- F3 weekly count in data/ is 11; we did not fabricate a missing row.

Version: v2 (post-Week-13 portal results).